TEXORA · LECTURE SHEET

Melt Spinning, Wet Spinning and Dry Spinning

Man-made filaments are formed by extruding a polymer liquid through a spinneret and solidifying the streams — by cooling (melt spinning), by evaporation (dry spinning) or by coagulation in a bath (wet spinning). This sheet compares the three routes and works their characteristic calculations: filament denier from pump rate, draw ratio, and solvent evaporation load.

YM-3207 · ADVANCED SPINNING CALCULATIONS · TOPIC 7 · SPECIALIZATION — YARN MANUFACTURING

1. One idea, three solidification routes

All filament spinning has the same skeleton: prepare a spinnable liquid, meter it through spinneret holes with a precision pump, solidify the extruded streams, draw them to develop strength, and wind up or collect as tow. The routes differ only in how the liquid is made and how it solidifies — and that difference decides which polymers each route can serve.

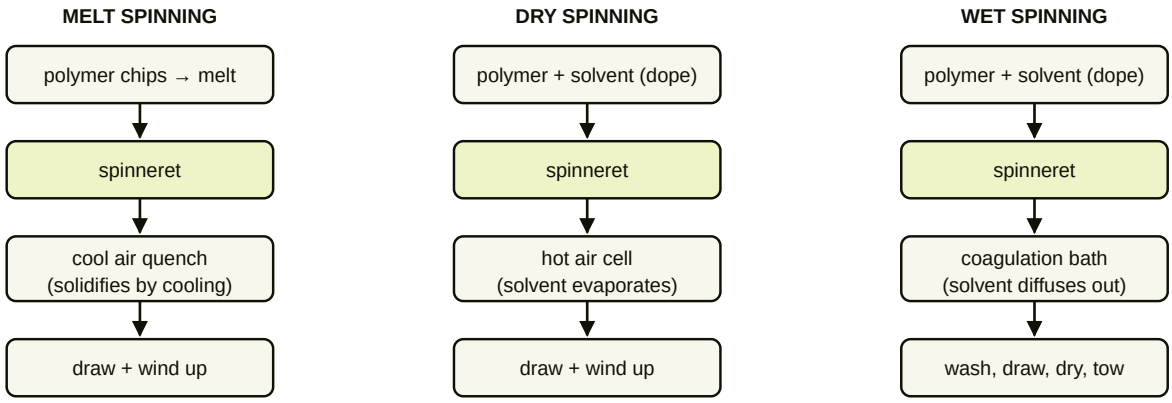


Figure 1 — The three filament-forming routes. Only the solidification step differs: cooling, evaporation, or coagulation.

2. Melt spinning and the denier calculation

Thermoplastic polymers that melt without decomposing — polyester, nylon, polypropylene — are melted, filtered and pumped through the spinneret; the streams solidify in a cool air quench and are

drawn to orient the molecules. No solvent is involved, so speeds are high (take-up 3000 m/min and above in modern plants) and the economics are the best of the three routes.

Filament size follows from mass conservation: what the pump pushes out per hole must equal what the take-up carries away.

$$\text{Denier per filament} = \text{pump rate per hole (g/min)} \times 9000 / \text{take-up speed (m/min)}$$

WORKED EXAMPLE — FILAMENT DENIER AND DRAW RATIO

A metering pump delivers 30 cm³/min of nylon melt (density 1.14 g/cm³) to a spinneret of 36 holes; as-spun filaments are collected at 600 m/min and then drawn at a draw ratio of 5.
Total melt rate = 30 × 1.14 = 34.2 g/min. Per hole = 34.2 / 36 = **0.95 g/min**.
As-spun denier = 0.95 × 9000 / 600 = **14.25 denier** per filament.
Drawing to ratio 5 reduces denier in proportion: final denier = 14.25 / 5 = **2.85 denier**.
Cross-check at the drawn take-up speed (600 × 5 = 3000 m/min): denier = 0.95 × 9000 / 3000 = **2.85** ✓. The two routes to the number agree, as they must — drawing changes length and denier together.

3. Dry spinning and the evaporation load

Polymers that decompose before melting but dissolve in a volatile solvent — cellulose acetate in acetone is the classic case — are dry spun: the dope is extruded into a heated cell, the solvent evaporates from the streams and is recovered, and the filaments solidify as the solvent leaves. The calculation that sizes the plant is the solvent balance.

WORKED EXAMPLE — SOLVENT EVAPORATION IN DRY SPINNING

A dry-spinning line produces 100 kg/h of acetate fibre from a dope containing 22% polymer.
Dope flow = 100 / 0.22 = **454.5 kg/h**.
Solvent to evaporate (and recover) = 454.5 − 100 = **354.5 kg/h**.
Every kilogram of fibre therefore drags about 3.5 kg of solvent through evaporation and recovery — the reason dry spinning is slower and costlier than melt spinning, and why solvent recovery efficiency dominates its economics.

4. Wet spinning and tow production

Wet spinning extrudes the dope into a liquid bath that is a non-solvent for the polymer: solvent diffuses out, non-solvent diffuses in, and the filament coagulates. Viscose rayon, most acrylic and some spandex are wet spun. Speeds are low (tens of metres per minute) because diffusion is slow, but the route

accepts polymers no other route can handle, and it produces staple fibre economically as a continuous **tow** that is washed, drawn, dried and cut.

WORKED EXAMPLE — WET-SPUN TOW PRODUCTION

A viscose tow consists of 100,000 filaments of 1.5 denier each, collected at 50 m/min.

Tow denier = 100,000 × 1.5 = 150,000 denier = 150,000/9000 = **16.67 g/m**.

Production = 16.67 × 50 = 833 g/min = **50.0 kg/h**.

The huge filament count compensates for the low speed: wet spinning wins on total mass throughput per machine, not on metres per minute.

5. The three routes compared

FEATURE	MELT	DRY	WET
Spinning liquid	Polymer melt	Solution in volatile solvent	Solution (dope)
Solidification	Cooling in air	Solvent evaporation in hot air	Coagulation in a bath
Typical polymers	Polyester, nylon, polypropylene	Acetate, some acrylic	Viscose, acrylic, spandex
Take-up speed	Very high (3000+ m/min)	Moderate	Low (tens of m/min)
Solvent recovery	None needed	Essential, large load (3.5 kg per kg fibre in example)	Bath regeneration needed
Typical product	Filament yarn, some staple	Filament yarn, tow	Staple fibre (tow), some filament
Relative cost	Lowest	High	Moderate–high

Glossary

Spinneret — the plate with fine holes through which the spinning liquid is extruded. **Dope** — the polymer solution prepared for dry or wet spinning. **Quench** — the cooling air that solidifies melt-spun filaments. **Draw ratio** — the ratio of final to initial length (or speed) in drawing; it orients molecules and raises strength. **Coagulation bath** — the non-solvent bath that solidifies wet-spun filaments by exchange of solvent and non-solvent. **Tow** — a large bundle of parallel filaments produced for cutting into staple fibre. **Metering pump** — the precision gear pump that fixes flow per spinneret hole.

Key points

• The three routes share one skeleton; only solidification differs — cooling, evaporation, coagulation.
• Melt spinning serves meltable polymers at the highest speeds and lowest cost; no solvent is involved.
• Denier per filament = pump rate per hole × 9000 / take-up speed; drawing divides denier by the draw ratio (14.25 → 2.85 in the example).
• Dry spinning's burden is solvent: 354.5 kg/h evaporated for 100 kg/h of fibre at 22% dope solids.
• Wet spinning is slow per filament (50 m/min) but massive in parallel — a 100,000-filament tow yields 50 kg/h.

Practice

1. A pump delivers 45 cm^3/min of polyester melt (density 1.20 g/cm^3) to a 48-hole spinneret with take-up at 3200 m/min (the filaments are drawn in-line, so this is the final denier). Calculate the denier per filament.
2. A dry-spinning dope contains 25% polymer and the line makes 80 kg/h of fibre. Calculate the dope flow and the solvent evaporation load.
3. An acrylic tow of 120,000 filaments of 1.3 denier is collected at 45 m/min. Calculate the production in kg/h.

Answer key

1. Total melt = 45 × 1.20 = 54 g/min; per hole = 54/48 = 1.125 g/min. Denier = 1.125 × 9000 / 3200 = **3.16 denier**.
2. Dope flow = 80 / 0.25 = **320 kg/h**. Solvent = 320 − 80 = **240 kg/h**.
3. Tow = 120,000 × 1.3 = 156,000 denier = 17.33 g/m. Production = 17.33 × 45 × 60 / 1000 = **46.8 kg/h**.